BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 140 · Lesson 140 · Referential Integrity, Indexing and Keys

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is the result of UPDATE without a WHERE clause?

Q2. What is a trace table used for?

Q3. How many bits are in one byte?

Q4. What data structure uses LIFO?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

referential integrity — Foreign-key values correspond to existing referenced keys or permitted NULLs.

index — Auxiliary structure helping the DBMS find rows efficiently.

primary key — Unique identifier of each record.

foreign key — Attribute referencing another relation.

New learning

Link the records — Order.CustomerID references Customer.CustomerID.

Enforce integrity — Invalid FK values are rejected unless permitted rules apply.

Improve searches — An index can speed lookup at the cost of extra storage/write overhead.

Worked example

Customer has IDs 7 and 9. A new Order has CustomerID 12. Explain what happens under enforced referential integrity.

1. Customers 7 and 9 — Identify the relevant information and make a prediction.

2. Order requests ID 12 — Explain the change using correct technical terms.

3. Reject orphan reference — The DBMS rejects the foreign key value 12, unless a matching Customer row is created or the reference is changed. Policies for deleting parent rows may RESTRICT, CASCADE or SET NULL, where permitted and configured.

We Do · Practise with guidance

A large Book table is queried by ISBN many times per day. Should the ISBN have an index?

Frequent ISBN lookup

Identify the relevant information and make a prediction.

Indexed search path

Explain the change using correct technical terms.

Write and space overhead

An index on ISBN can support quicker exact-match retrieval, especially for large data, but occupies space and requires maintenance after insert/update/delete. A UNIQUE constraint is appropriate if ISBN must be unique in this table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain referential integrity using Borrower and Loan with a suitable example. [6]

Question 2

Identify two benefits and one cost of indexing a frequently searched field. [6]

Question 3

A library deletes Borrower ID 8 while Loan rows refer to 8. Compare RESTRICT and CASCADE and justify a safe policy. [8]

Exit Ticket · Answer before checking

An index guarantees every column value it indexes is unique.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Every row in the target table can be updated.

Q2. Tracking variable state changes as an algorithm runs.

Q3. Eight.

Q4. Stack.

You Do pages · Independent questions

Q1. Loan FK references existing Borrower key [2], prevents dangling child rows [2], example insertion/deletion rule [2].

Q2. Faster key lookups [2], faster ordered/range operations where supported [2], storage or index-update overhead [2].

Q3. RESTRICT denies parent deletion [2]; CASCADE removes dependent loans [2]; loss/history risks [2]; reasoned chosen retention/archival policy [2].

Exit Ticket · Model answer

False. Uniqueness requires a unique index or constraint.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: